



Gun Tank Sulfur Hexafluoride Handling Procedure

1.7.27.2 Rev 2

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REVISION HISTORY

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0	2002-Feb-17	Comments from Mohamed Benmerrouche, Allen Hodges, Bob Crosby and Johan Greefkes	Neil Johnson
0A	2005 January 19	Additional comments from following procedure.	Neil Johnson
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1.0 SCOPE

Sulfur Hexafluoride (SF₆) is used as an insulator in the gun tank. A current MSDS sheet is attached at the end of this procedure. This procedure describes the steps required to transfer the SF₆ from the gun tank to the storage tanks, and how to fill the gun tank from the storage tank.

2.0 QUALIFICATIONS

Two people are required when transferring SF₆. Operation of the SF₆ pumping system shall be supervised by a lead person trained in its operation and familiar with this document, the MSDS in the appendices and the hazards associated with SF₆ gas. An untrained assistant can assist this supervisor and will be encouraged to get hands-on experience operating the equipment. Two sessions of hands-on training involving emptying and refilling the gun tank are sufficient to meet the training requirements of this section. For the second session the trainee shall follow this procedure and operate the controls, without assistance from the supervisor. The gun tank area is a radiological controlled area, so at least one person must have Radiological Worker Training (RWT). The CLS HSE department is responsible for maintaining a record of employees who are qualified as a lead person and who have had on the job training while being supervised.

3.0 HAZARDS

3.1 ELECTRICAL

The gun operates with a high voltage of 220 kV. High voltages of up to a few kV are also present on the hot deck inside the gun. Circuit breakers on the control power and magnetics panel P1021.1-07 and the electron gun HVPS control equipment panel P1021.1-05 must be switched off. The circuit breaker for the Haimson console in DPNL1021-must be switched off, tagged and locked out. The power connector at the gun high voltage power supply tank should be disconnected.

3.2 PHYSICAL

A leak on the input side of the transfer pump could result in an overpressure on either the storage tanks or the gun tank. If the pressure relief fails, the tank could burst with a potential for flying debris. The shock wave could cause hearing damage. The tank pressures must be monitored while the pump is running.

Note: If the SF₆ is contaminated with air, it will be replaced.

3.3 TOXICITY

If sulfur hexafluoride is subjected to electrical discharge it can form highly toxic decomposition products. When exposed to flame or electric arc SF₆ can decompose to the toxic oxides SOF₂ and sulfuryl fluoride SO₂F₂. The LC₅₀ concentration (concentration lethal to 50% of population) for SO₂F₂ is 1200 to 3000 ppm for one hour^[1]. Other possible hazardous decomposition products include fluorinated compounds, oxides of sulfur, sulfur compounds, hydrogen fluoride, and hydrogen sulfide. The gun tank shall be well ventilated using a fan (> 20 cm diameter) before working in it. If the SF₆ has been exposed to extensive burning or arcing it should be discarded.

3.4 EQUIPMENT DAMAGE/FIRE

Lens 1 SOL0001-01 and the gun steering magnets ST0001-01 is located inside the gun tank. They require convection cooling, normally by the SF₆. These will overheat and may be damaged if they are energized when the gun tank is evacuated. Power must be removed as described in Section 4.1 before evacuating the tank.

4.0 TRANSFERRING SF₆ FROM GUN TANK TO STORAGE TANKS

4.1 DISCONNECT POWER TO GUN TANK

Switch off and lock out the breaker on the control power and magnetics panel P1021.1-07.

Switch off and lock out the breaker on the electron gun HVPS control equipment panel P1021.1-05 Circuit 22.

Switch off, lock and tag out the circuit breaker for the Haimson console in DPNL1021-01 following CLS LOTO procedure 11.7.37.1 Rev 4.

Disconnect the power connector at the gun high voltage power supply tank.

4.2 EVACUATE TRANSFER LINES AND STORAGE TANKS

Observe that transfer pump is not running.

Observe that the valve at the SF₆ regulator output is closed.

If the valve at the SF₆ regulator output is not closed, purge transfer piping up to valve with SF₆ as follows:

- Verify the vacuum pump is not connected

- Adjust valve at the SF₆ regulator output to slightly open position.

- Set VACUUM PUMP VALVE to "OPEN" position.

- Open the valve on the SF₆ gas bottle.

- Observe that SF₆ can be heard flowing through system.

- Close valve at SF₆ regulator output.

- Close valve on SF₆ gas bottle.

- Set VACUUM PUMP VALVE to "CLOSED" position.

Connect portable vacuum pump to outlet on gas transfer control valve panel.

Turn vacuum pump on.

Set VACUUM PUMP VALVE to "OPEN" position.

Set BYPASS valve to "OPEN" position.

Set SF₆ CYL. Valve to "OPEN" position.

Slowly open valves on each storage tank.

Wait until STORAGE TANK pressure gauge stops falling and reads approximately -30 PSI.

Cycle pump valves from "CLOSED" to "FROM GUN TANK TO STORAGE TANK" to "FROM STORAGE TANK TO GUN TANK" and back to "CLOSED" to evacuate any trapped air.

Wait at least 5 minutes.

Verify that the pressure has not changed.

Set SF₆ CYL. Valve to "CLOSED" position.

Set VACUUM PUMP valve to "CLOSED" position.

Turn vacuum pump off.

4.3 TRANSFER SF₆ FROM GUN TANK TO STORAGE TANKS

Verify BYPASS valve is in "OPEN" position.

Verify VACUUM PUMP valve is in CLOSED position.

Remove brass cover from valve at gun tank.

Adjust valve at gun tank to be slightly open.

Observe that SF₆ can be heard flowing through system.

Wait for pressure in gun tank and storage tanks to equalize. The valve at the gun tank can be opened more fully as flow rate decreases.

Set the valve at the gun tank to fully open.

Set the BYPASS valve to the CLOSED position.

Set PUMP valves to "FROM GUN TANK TO STORAGE TANK" positions.

Press green START button to start transfer pump.

Observe regular pulsation of STORAGE TANK pressure gauge. Their amplitude will decrease as gun tank pressure drops.

Monitor STORAGE TANK pressure gauge. If it goes much above 60 PSI there is probably an air leak on the input side of the pump, which must be investigated. Press the red STOP button to stop the transfer pump, close all valves, and if necessary seek assistance to locate the cause of the problem.

Wait until GUN TANK pressure gauge stops falling and reads approximately -30 PSI.

Wait at least 5 minutes more.

Verify that the pressure has not changed.

Press the red STOP button to stop the transfer pump.

Set PUMP valves to "CLOSED" position.

Close valves on SF₆ storage tanks.

4.4 OPEN GUN TANK

Disconnect vacuum pump if it is still connected.

Gradually open VACUUM PUMP valve to fill the gun tank with air.

Unbolt and remove the gun access port cover.

Place a fan on a table or other support and direct its flow into the tank.

Wait 5 minutes to thoroughly flush the tank with fresh air.

5.0 TRANSFERRING SF₆ FROM STORAGE TANKS TO GUN TANK

5.1 DISCONNECT POWER TO GUN TANK

Switch off the breaker on the control power and magnetics panel P1021.1-07.

Switch off the breaker on the electron gun HVPS control equipment panel P1021.1-05

Switch off, tag and lock out the circuit breaker for the Haimson console in DPNL1021-01.

Disconnect the power connector at the gun high voltage power supply tank.

5.2 EVACUATE TRANSFER LINES AND GUN TANK

Observe that transfer pump is not running.

Observe that the valve at the SF₆ regulator output is closed.

If the valve at the SF₆ regulator output is not closed, purge transfer piping up to valve with SF₆ as follows:

- Verify the vacuum pump is not connected

- Adjust valve at the SF₆ regulator output to slightly open position.

- Set VACUUM PUMP VALVE to "OPEN" position.

- Open the valve on the SF₆ gas bottle.

- Observe that SF₆ can be heard flowing through system.

- Close valve at SF₆ regulator output

- Close valve on SF₆ gas bottle

- Set VACUUM PUMP VALVE to "CLOSED" position.

Verify the valve at the gun tank is fully open.

Connect portable vacuum pump to outlet on gas transfer control valve panel.

Turn vacuum pump on.

Set VACUUM PUMP VALVE to "OPEN" position.

Set BYPASS valve to "OPEN" position.

Set SF₆ CYL. valve to "OPEN" position.

Wait until STORAGE TANK pressure gauge stops falling and reads approximately -30 PSI.

Cycle pump valves from "CLOSED" to "FROM GUN TANK TO STORAGE TANK" to "FROM STORAGE TANK TO GUN TANK" and back to "CLOSED" to evacuate any trapped air.

Wait at least 5 minutes more.

Set SF₆ CYL. valve to "CLOSED" position.

Set VACUUM PUMP valve to "CLOSED" position.

Turn vacuum pump off.

5.3 TRANSFER SF₆ FROM STORAGE TANKS TO GUN TANK

Verify BYPASS valve is in "OPEN" position.

Verify VACUUM PUMP valve is in CLOSED position.

Slowly open valves on storage tanks.

Observe that SF₆ can be heard flowing through system.

Wait for pressure in gun tank and storage tanks to equalize.

Set the BYPASS valve to the CLOSED position.

Set PUMP valves to "FROM STORAGE TANK TO GUN TANK" positions.

Press green START button to start transfer pump.

Observe regular pulsation of GUN TANK pressure gauge. Their amplitude will decrease as storage tank pressure drops.

Monitor GUN TANK pressure gauge. If it goes above 50 PSI there is probably an air leak on the input side of the pump, which must be investigated. Press the red STOP button to stop the transfer pump, close all valves, and seek assistance.

Wait until STORAGE TANK pressure gauge reads -30 PSI.

Close valves on SF₆ storage tanks.

If the gun tank pressure is below 50 PSI:

- Slowly set SF₆ CYL. valve to "OPEN" position.

- Open valve on SF₆ gas bottle

- Slowly open valve after SF₆ regulator.

- Observe that the pump sound changes. It may be necessary to adjust the pressure regulator at the SF₆ bottle to get a slow flow.

- When GUN TANK pressure gauge reaches 50 PSI close valve after SF₆ regulator:

 - Close valve on SF₆ gas bottle.

 - Set SF₆ CYL VALVE to "CLOSED" position.

Press the red STOP button to stop the transfer pump.

Set PUMP valves to "CLOSED" position.

Close valve at gun tank.

Disconnect vacuum pump.

Notify the monitoring personnel in the control room that the procedure is finished.

Connect power cable to gun tank.

Remove lock out tags and switch breakers on.

6.0 DISPOSAL

Contaminated sulfur hexafluoride is left in the storage tanks and sent out for disposal. The gun tank can be completely refilled using new gas from the factory bottle.

Saskatchewan Environment classifies sulfur hexafluoride as a hazardous material. It must be disposed of according to federal, provincial, and municipal regulations.

7.0 APPENDICES

7.1 APPENDIX 1 – SULFUR HEXAFLUORIDE MSDS

[http://www.praxair.com/praxair.nsf/AllContent/C21201459EDDBCA285256A86008221A2/\\$File/p4657d.pdf](http://www.praxair.com/praxair.nsf/AllContent/C21201459EDDBCA285256A86008221A2/$File/p4657d.pdf)

7.2 APPENDIX 2 – SULFURYL FLUORIDE MSDS

<http://www.cdms.net/LDat/mp5KO001.pdf>

8.0 REFERENCES

1. Effects of Exposure to Toxic Gases First Aid & Medical Treatment, Matheson Gas Products.